

Suraj Kumar

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EDUCATION

Case Western Reserve University (CWRU)

Cleveland, OH

Master of Science (M.S.), Computer Science

Aug 2026 - Dec 2027

Bachelor of Science in Engineering (B.S.E.), Biomedical Engineering

Aug 2023 - Dec 2027

Relevant Coursework: Operating Systems (C), Intro to AI, Algorithms, Data Structures (Java), Discrete Math

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, HTML/CSS, SQL, Java, C

Frameworks & Libraries: Django, FastAPI, React.js, React Native, Next.js, PyTorch, TensorFlow, Scikit-learn, vLLM, HuggingFace, OpenCV, SQLAlchemy, Selenium

Tools: Git, AWS (ECS, VPC, RDS, S3), Docker, PostgreSQL, Linux, REST APIs, CI/CD, Gemini API, Claude Code

SOFTWARE PROJECTS

AcoustiCare | Software Engineering Lead | *AWS, Docker, Django, React Native, Expo EAS* Nov 2025 – Present

- Led development of a voice-screening mobile app for ENT physicians that automates audio analysis of patient vocal health, **cutting diagnostic time from 2+ hours to seconds** per patient.
- Architected scalable AWS infrastructure (VPC, ECS, RDS PostgreSQL, S3, KMS) supporting **100+ users**, with layered security groups and TLS-terminated load balancing to store sensitive clinical data at scale.
- Drafted a provisional patent white paper and established a clinical data collection pipeline with ENT physicians at UH Hospital to validate the system on real patient data.

Rival | Software Engineering Lead | *React Native, FastAPI, Gemini API, OAuth 2.0 (GitHub, Notion)* Jan 2026

- Built a GenAI-powered 1v1 mobile game using Gemini API with structured prompt engineering to score GitHub and Notion contributions via OAuth-integrated APIs, automatically determining who was more productive.
- Engineered the full-stack system (challenge UI, real-time progress tracking, and AI verdict pipeline) to handle concurrent multiplayer matches end to end.

WORK/RESEARCH EXPERIENCE

AI Security Graduate Researcher

May 2026 – Present

Case Western Reserve University PEAT AI Lab

Remote/Cleveland, OH

- Built an unsupervised, online log-anomaly detection framework that flags intrusions in streaming network-flow logs at **0.99 AUROC**, scoring each window against a learned geometric typicality region.
- Engineered TF-IDF, Word2Vec, and Qwen3-based embedders paired with four outlier detection models (GMM, KDE, One-Class SVM, DeepSVDD) to convert raw log streams into actionable anomaly alerts.
- Scaled embedding generation on Linux HPC clusters via chunked streaming I/O and memory-efficient attention, processing **10M+ flow records** while **cutting peak GPU memory by 10x**.

Machine Learning Engineer Intern

Jun 2025 – Dec 2025

Genentech

South San Francisco, CA

- Reduced the Bioassay department's experimental workload by 30%** by training machine learning models and neural networks (PyTorch and Scikit-learn) to predict drug-potency experiment failures.
- Deployed ML inference endpoints to AWS EKS via GitLab CI/CD pipelines with **100%** automated test coverage.
- Optimized SQL queries across 30+ relational tables, **cutting data retrieval from 2 hours to 10 minutes**.
- Drove adoption of the analytics tool across **3 departments and 100+ employees** through technical demos to engineers, scientists, and stakeholders.

Deep Learning Undergraduate Researcher

Sep 2024 – Nov 2025

Case Western Reserve University INVent Lab

Remote/Cleveland, OH

- Engineered a deep learning pipeline for glomeruli segmentation in pathology images that leverages a foundation model (MedSAM) to provide pseudo-labels for a UNet, **increasing baseline accuracy by 19%**.
- Trained an EfficientNet-B3 binary classifier to filter non-glomeruli tiles at **94% precision**, reducing downstream segmentation noise and **shrinking the testing pool by 40%** before MedSAM labeling.
- Refined noisy ground truth masks via morphological opening and connected-component area filtering, discarding spurious segmentations to raise pseudo-label quality across training iterations.